

ORIGINAL RESEARCH

Perfluoroalkyl/Polyfluoroalkyl Substances, Hemostatic/Inflammatory Biomarkers, and Incident Hypertension



The Study of Women's Health Across the Nation

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ABSTRACT

BACKGROUND Perfluoroalkyl/polyfluoroalkyl substances (PFAS) are emerging clinical and public health concerns, but their role in cardiovascular disease risk remains unclear. Inflammation and thrombosis may link PFAS and cardiovascular disease.

OBJECTIVE We investigated the association between serum PFAS concentrations and inflammatory and hemostatic markers in midlife women, and whether these markers mediated the association between PFAS and incident hypertension.

METHODS We analyzed 1,387 participants from the Study of Women's Health Across the Nation from 1999 to 2000 to 2015 to 2016. Linear mixed-effects models estimated the association of 7 PFAS with high-sensitivity C-reactive protein (hs-CRP), fibrinogen, factor VII activity, plasminogen activator inhibitor-1 (PAI-1), and tissue plasminogen activator antigen. Quantile-g-computation evaluated mixture effects. Causal mediation analyses tested whether these markers mediated the PFAS-hypertension association.

RESULTS Individual PFAS were positively associated with hs-CRP and PAI-1. For hs-CRP, per doubling of PFAS ranged from 3.59% (95% CI: 0.01%-7.30%) for branched perfluorooctane sulfonic acid isomers to 5.32% (95% CI: 1.67%-9.10%) for 2-(N-methyl-perfluorooctane sulfonamido) acetic acid. For PAI-1, per doubling of PFAS ranged from 2.41% (95% CI: -0.86% to 5.78%) for the linear perfluorooctane sulfonic acid isomer to 3.04% (95% CI: 0.24%-5.93%) for 2-(N-methyl-perfluorooctane sulfonamido) acetic acid. However, none of these associations remained statistically significant after false discovery rate correction. PFAS mixture effects were 6.15% (95% CI: -0.83% to 13.58%) for hs-CRP and 6.05% (95% CI: 0.29%-12.14%) for PAI-1. No associations were found with other markers, and biomarkers did not mediate the PFAS-hypertension association.

CONCLUSIONS These findings suggest, but do not establish, that PFAS may be linked to inflammation and impaired fibrinolysis in midlife women. (JACC Adv. 2026;5:103071) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

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**ABBREVIATIONS
AND ACRONYMS****FVIIc** = factor VII activity**hs-CRP** = high-sensitivity
C-reactive protein**MeFOSAA** = 2-(N-methyl-
perfluorooctane sulfonamido)
acetic acid**n-PFOS** = linear
perfluorooctane sulfonic acid
isomer**PAI-1** = plasminogen activator
inhibitor-1**PFAS** = perfluoroalkyl and
polyfluoroalkyl substances**Sm-PFOS** = branched
perfluorooctane sulfonic acid
isomers**SWAN** = Study of Women's
Health Across the Nation**tPA-antigen** = tissue
plasminogen activator antigen

Perfluoroalkyl and polyfluoroalkyl substances (PFAS) are synthetic chemicals extensively used in various consumer products and industrial applications because of their oil-repellent, water- and stain-resistant properties. The environmental persistent nature of these chemicals has resulted in a rise in environmental pollution, with consequent negative impacts on human health.¹ PFAS do not break down, and thus, many bioaccumulate in human tissues, such as liver, bones, lungs, kidneys, and brain, posing potential health risks.² Despite recent reductions in blood serum concentrations of legacy PFAS, such as perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), the estimated number of PFAS as a chemical class globally is around 15,000,³ raising concerns due to the lack of comprehensive studies on their toxicities. Available epidemiologic data suggests that PFAS exposure in the general

population may be associated with decreased fertility, preeclampsia, increased cholesterol levels, risk of obesity, reduced immune functions, and some cancers including prostate, kidney, and testicular cancers.⁴ PFAS also have been associated with increased concentrations of some inflammatory biomarkers such as interleukin (IL)-1 β , IL-6, and tumor necrosis factor- α , indicating the potential effect of PFAS on the inflammatory pathways.⁵⁻⁷

Inflammation and hemostasis play a pivotal role in the pathophysiology of chronic conditions. C-reactive protein (CRP), a hepatic pentameric protein, acts as a systemic marker of inflammation, with elevated levels often indicating acute infection or persistent inflammatory diseases such as rheumatoid arthritis.⁸ Hemostatic markers—which regulate blood coagulation and fibrinolysis—are activated during injury or inflammation by proinflammatory agents, including tissue factors and cytokines.⁹ Hemostatic markers are important indicators of cardiovascular disease (CVD)¹⁰ and are related to CVD subclinical diseases like pulmonary hypertension.¹¹ Fibrinogen is a key coagulation component that forms clots to stop bleeding, whereas factor VII coagulant activity (FVIIc) provides an index of coagulation velocity.⁹ Fibrinolysis, the counterbalance to clot formation, is modulated by the interplay of the tissue plasminogen

activator (tPA) and its inhibitor (e.g., plasminogen activator inhibitor-1 [PAI-1]).⁹ Chronic inflammation could influence hemostasis balance, enhancing coagulation and impairing fibrinolysis, thereby potentiating CVD development.¹²

While CRP and hemostatic markers serve as important indicators of inflammation and CVD, epidemiologic research into the relationship between PFAS exposure and systemic inflammation as well as hemostatic markers has been limited and yielded mixed results. Only a few epidemiologic studies have found a positive association between PFAS exposure and serum CRP level,¹³ whereas other studies have yielded null or contradictory results.^{7,14-17} Little is known about the association between PFAS exposure and hemostasis. Two in vitro studies suggested a potential link between PFAS exposure and increased blood coagulation activity.^{18,19} To the best of our knowledge, no epidemiologic studies have investigated the association between PFAS exposure and hemostatic markers.

Epidemiologic evidence also supports an association between PFAS exposure and increased risk of CVD.²⁰⁻²² A recent study from the SWAN (Study of Women's Health Across the Nation) found an association between serum PFAS concentrations and an elevated risk of developing hypertension.²⁰ However, little is known about the mediating roles of systemic inflammation and hemostasis in the relationship between PFAS exposure and CVD, which could help elucidate the underlying biological mechanisms.

We examined: 1) the association between serum concentrations of select PFAS, individually and as mixtures, and markers of systemic inflammation and hemostasis including high-sensitivity CRP (hs-CRP), fibrinogen, FVIIc, PAI-1, and tPA-antigen; and 2) whether hs-CRP and hemostatic markers mediated the association between PFAS and incident hypertension in a cohort of midlife women, SWAN. Midlife women are at elevated risk of developing CVD due to hormonal changes that occur in this life stage.²³ Identifying modifiable risk factors that are widespread across the population is critical. This study aims to advance our understanding of how widespread exposure to PFAS could influence the risk of CVD by examining inflammatory and hemostatic markers.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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METHODS

STUDY POPULATION. SWAN is a longitudinal, multi-racial/ethnic study aimed at understanding the physiological and psychosocial changes during the menopause transition and their impact on the risk factors for chronic diseases associated with aging.²⁴ In 1996 to 1997, 3,302 premenopausal women, aged between 42 and 52 years, were enrolled from 7 locations across the United States, including Boston, MA; Chicago, IL; Southeast Michigan (Ypsilanti, Inkster), MI; Los Angeles, CA; Newark, NJ; Oakland, CA; and Pittsburgh, PA. Eligible participants had an intact uterus and at least one ovary, at least one menstrual cycle in the past 3 months, and not have used exogenous hormone in the last 3 months. The women identified themselves as either non-Hispanic White or as belonging to one of the designated minority groups from each site, including Black (Chicago, Southeast Michigan, Pittsburgh, Boston), Chinese (Oakland), Japanese (Los Angeles), or Hispanic (Newark). Clinical examination data and biological samples were collected nearly annually. The study was approved by the institutional review board at each site, and all participants provided written informed consent at each study visit.

In 2016, the SWAN Multi-Pollutant Study (MPS) was launched, using the third SWAN follow-up visit (V03, 1999-2000) as its baseline to investigate the health effects of exposure to various environmental chemicals, including PFAS, among middle-aged women. Hereafter, “baseline” refers to SWAN follow-up visit 03. The MPS utilized existing serum and urine samples for environmental exposure assessments. After excluding participants from Chicago (N = 368) and Newark (N = 278) due to the unavailability of urine samples and other participants due to insufficient sample volumes (N = 648), the final MPS cohort consisted of 1,400 women.

In the SWAN study, PAI-1 and tPA-antigen measurements were collected from V03 through the seventh follow-up visit (V07, 2003-2005). Measurements for fibrinogen and FVIIc were taken at V03, V05, and V07. At V07, these markers were obtained from a randomly selected half of the participants. Although fibrinogen and FVIIc were also measured at the fifteenth follow-up visit (V15, 2015-2016), there was a 12-year gap in assessment of these biomarkers after V07. Therefore, we included hemostatic markers from V03 to V07 only. hs-CRP was measured at V03-V07, V09 (2005-2006), V10 (2006-2007), V12 (2009-2011), and V15 (2015-2016).

We included all available hs-CRP observations from V03 to V15.

Of the original 1,400 SWAN MPS participants, we excluded 9 who were missing core covariates, such as body mass index, smoking status, and menopausal status at baseline and participants with missing or flagged values for inflammatory or hemostatic biomarkers. Thus, the analytic sample varies slightly depending on the biomarker of interest. For hs-CRP, the analytic sample includes 1,342 participants contributing a total of 10,640 observations; for PAI-1 and tPA-antigen, the analytic sample includes 1,226 participants contributing 6,229 observations; and for fibrinogen and FVIIc, 1,239 participants and 3,086 observations are in the analytic sample. For the mediation analyses, we further excluded 290 women with prevalent hypertension at visit 3 for hs-CRP, PAI-1, and tPA, and 317 women with prevalent hypertension at visit 4 for fibrinogen and factor VIIc, so that only incident hypertension cases contributed to the mediation analyses. The sampling processes for each biomarker are illustrated in [Supplemental Figure 1](#).

MEASUREMENT OF SERUM PFAS. Baseline serum samples from SWAN MPS were sent to the Division of Laboratory Sciences at the Centers for Disease Control and Prevention for PFAS assessment. The analysis of de-identified specimens at the Centers for Disease Control and Prevention laboratory was determined not to constitute engagement in human subjects research. Eleven PFAS were measured including perfluorohexane sulfonic acid (PFHxS), linear PFOS isomer (n-PFOS), branched PFOS isomers (Sm-PFOS), linear PFOA isomer (n-PFOA), branched PFOA isomers (Sb-PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), 2-(N-methyl-perfluorooctane sulfonamido) acetic acid (MeFO-SAA), and 2-(N-ethyl-perfluorooctane sulfonamido) acetic acid (EtFOSAA) using online solid-phase-extraction high-performance liquid-chromatography isotope-dilution tandem mass spectrometry.²⁵ Detailed descriptions of the analytical techniques and quality control measures are documented elsewhere.^{26,27} The quality control was reflected in the consistency of our measurements, with the coefficient of variation (CV) for the low- and high-concentration quality control pools ranging between 5.9% and 12.1%. The limit of detection (LOD) for all PFAS tested was 0.1 ng/mL. Concentrations below the LODs were imputed as LOD/ $\sqrt{2}$. Sb-PFOA, PFDA, PFUnDA, and PFDoDA were excluded from

data analyses because of their relatively low detection frequency.

MEASUREMENT OF HS-CRP AND HEMOSTATIC BIOMARKERS. Plasma hs-CRP levels were measured using an ultrasensitive immunonephelometric method (Dade-Behring, Marburg, Germany) that achieved a sensitivity threshold of 0.03 mg/dL. The variability between assays represented by the CV ranged from 10% to 12% for the low-concentration hs-CRP (0.05 mg/dL) and 5% to 7% for the high-concentration hs-CRP (2.2 mg/dL).²⁸ The potential acute inflammation cutoff point for hs-CRP value was set to 10 mg/dL due to the varied ethnicities and body weight among participants.²⁹ hs-CRP values larger than 100 mg/L were flagged. A total of 6 hs-CRP values were excluded because they were >100 mg/L. Measurements for fibrinogen and FVIIc were conducted on thawed citrated plasma samples using a turbidometric method with the MLA ELECTRA 1400C instrument (Medical Laboratory Automation Inc.). Fibrinogen monthly CVs varied between 2.3% and 3.5% for the average concentration of 250-mg/dL pool and 2.6% to 3.6% for the 140-mg/dL pool, while FVIIc showed monthly CVs of 7.8%, 5%, and 4% at mean activity levels of 8%, 45%, and 99%, respectively. Fibrinogen values were flagged if they were <128 mg/dL or >1,020 mg/dL; however, no fibrinogen values were flagged. For FVIIc, values were flagged if they fell below 12.5% or exceeded 200%; consequently, 80 FVIIc values were excluded due to exceeding 200%. PAI-1 concentrations were determined through a sandwich ELISA technique employing a solid-phase monoclonal antibody and an enzyme-labeled goat antiserum kit (American Diagnostica). The PAI-1 assay showed monthly CVs of 5% to 9% at an average level of 7 ng/mL and 4% to 9% at 22.5 ng/mL. Values were flagged if they were ≥ 150 ng/mL or <0 ng/mL, leading to the exclusion of 78 PAI-I values due to exceeding the 150 ng/mL threshold. tPA-antigen levels were quantified using a double-antibody ELISA technique (American Diagnostica) and were calibrated using human single-chain tPA (NIBSAC) against an international standard. This assay's monthly CVs ranged from 4.7% to 8.7% at a mean concentration of 5.6 ng/dL, and from 3.8% to 7.8% at 11 ng/mL. Values <1 ng/mL or >30 ng/mL were flagged.³⁰ Consequently, 51 tPA-ag values were excluded because they exceeded 30 ng/mL, and 5 values were excluded because they were below 1 ng/mL. All flagged values identified as being outside the laboratory-defined ranges were excluded from the analysis, in accordance with SWAN data documentation. These flagged

values may reflect either assay artifacts or biological outliers unrelated to chronic exposure. Their exclusion was intended to minimize the impact of extreme measurements on the study findings.

INCIDENCE OF HYPERTENSION. Hypertension was defined by the measurement of both systolic blood pressure and diastolic blood pressure. At each follow-up visit, seated blood pressure was measured using a standardized protocol. After at least 5 minutes of rest, 2 consecutive readings were obtained on the right arm with an appropriately sized cuff, using an auscultatory method and a standard sphygmomanometer. Participants were instructed to avoid smoking and caffeinated beverages for at least 30 minutes before measurement. If systolic blood pressure is ≥ 130 mm Hg, diastolic blood pressure ≥ 80 mm Hg, or any current use of antihypertensive medication is reported, the patient was considered as having hypertension.³¹

COVARIATES. Covariates were selected based on prior knowledge. Sociodemographic variables included age (year), SWAN site, race and ethnicity (White, Black, Chinese, Japanese), and education (less than high school, high school, college, post college). Other time-invariant health-related variables at SWAN MPS baseline included body mass index (kg/m²), nonsteroidal anti-inflammatory drugs (NSAIDs) use (yes/no), and menopausal status. Menopausal status was based on menstrual bleeding patterns over the past year: premenopausal (menstrual periods in the last 3 months without a change in the pattern of bleeding), perimenopausal (less regular or absent menstrual periods in the last 3 months), surgery or natural postmenopause (no menstrual periods for 12+ months not due to hormone therapy or chemotherapy), and unknown (use of hormone therapy in the past year prior to establishing a date for the final menstrual period). These variables were treated as time invariant because PFAS may affect these variables; thus, adjustment for these variables measured after PFAS assessment could lead to overadjustment bias.³²

Time-dependent covariates included self-reported smoking status (never, former, current), passive smoking status (0 h/week, 1-4 h/week, 5+ hours/week), alcohol consumption (none, <1 serving/week, ≥ 1 serving/week), total daily calorie intake, and physical activity score. The total calorie intake was determined by adding up the energy content corresponding to the frequency of consumption for each food item, as reported in a comprehensive semiquantitative food frequency questionnaire.³³ Physical activity score was evaluated through a

modified version of the Kaiser Physical Activity Survey, with scores ranging from 3 to 15.³⁴

STATISTICAL ANALYSIS. Descriptive analyses were performed to summarize the characteristics of the study participants, PFAS, and inflammatory and hemostatic biomarker concentrations at baseline. All serum PFAS concentrations were log-2 transformed to capture potential log-linear relationships and for a better interpretation. All inflammatory and hemostatic biomarkers were natural-log transformed to ensure normality.

Linear mixed-effects models were employed to assess the longitudinal association between PFAS serum concentrations and all 5 biomarkers. Models included a participant-specific random intercept to account for within-person correlations. Fixed effects included PFAS, years since baseline, and prespecified covariates. We did not include random slopes for time. Effect estimates were converted back and presented as percentage differences and 95% CIs in inflammatory and hemostatic biomarker levels per doubling of PFAS concentrations. Given the number of PFAS-biomarker tests conducted, we used the Benjamini-Hochberg false discovery rate (FDR) procedure to adjust for multiple comparisons. Both unadjusted and FDR-adjusted *P* values were reported.

To evaluate the robustness of our findings, we conducted several sensitivity analyses. First, we additionally adjusted for NSAIDs use given the potential link between PFAS exposure and heightened inflammatory symptoms and conditions such as hypertension³⁵—which may lead to the use of such medications. Second, to assess whether associations varied over follow-up, we tested PFAS and time interaction. Third, we restricted hs-CRP observations to visits V03-V07 to harmonize the follow-up window with the fibrinogen and FVIIIc analyses and to assess whether longer follow-up influenced the hs-CRP estimates. Fourth, we assessed potential effect modification by race using formal multiplicative interaction analyses by adding PFAS-by-race interaction terms to the fully adjusted models, with race categorized as White, Black, and Asian (including Chinese and Japanese participants). Finally, we conducted a sensitivity analysis including biomarker observations with flagged extreme values to assess the impact of these outliers on the results.

To evaluate the joint effects of PFAS mixtures on inflammatory and hemostatic markers, we used quantile g-computation (qgcomp), a statistical approach designed to estimate the overall effect of exposure mixtures.³⁵ This method quantifies the change in the outcome associated with a

simultaneous increase in all PFAS mixtures by one quantile, accommodating exposures with effects in both positive and negative directions. In our analysis, we included the serum concentrations of 7 PFAS as exposure variables: n-PFOA, PFNA, PFHxS, n-PFOS, Sm-PFOS, MeFOSAA, and EtFOSAA. Total PFOS was not included because it is calculated as the sum of n-PFOS and Sm-PFOS, which were already part of the qgcomp analysis. These 7 PFAS concentrations were modeled in quartiles, consistent with common qgcomp practice, to provide a flexible parameterization that is less sensitive to extreme values. Each biomarker outcome was modeled separately, adjusting for the same covariates included in the linear mixed-effects model. Interaction terms among PFAS mixture components were not included because our primary estimand was the overall joint mixture effect under an additive-components assumption. Dependence of the repeated observations was handled by the option “id.” We applied the bootstrap method to calculate cluster-appropriate standard errors for our estimates. The bootstrap approach involved 2,000 iterations (*B* = 2,000) to derive CIs.

To investigate whether inflammatory and hemostatic markers mediated the associations between PFAS and incident hypertension, we conducted causal mediation analyses within an accelerated failure time (AFT) framework. Cox proportional hazards models, which require the rare outcome assumption,³⁶ were not used due to the high incidence of hypertension in this population. For each PFAS-mediator model, time from visit 3 to incident hypertension was modeled using Weibull AFT models with log₂ PFAS at visit 3 as the exposure and the log-transformed biomarker as the mediator. hs-CRP, PAI-1, and tPA were treated as mediators measured at visit 4, with follow-up for incident hypertension beginning after visit 4. Fibrinogen and factor VIIIc were modeled using visit 5 values, with follow-up beginning after visit 5. Outcome models included PFAS, the mediator, and the same covariates described earlier, including both linear and squared terms for age, consistent with prior SWAN PFAS analyses. PFAS concentrations were modeled linearly on the log₂ scale to estimate interpretable effects per doubling.²⁰ We used linear regression models as the mediator models. Two sets of models were estimated: one including only main effects of PFAS and the mediator, and a second including a PFAS times mediator product term. Because the exposure-mediator interaction was statistically significant for several PFAS-mediator combinations (two-sided *P* < 0.05), estimates from the interaction

models were considered as primary. Using the parametric g-formula implementation in the CMAverse package with 1,000 bootstrap resamples, we estimated the natural direct effect (effect of PFAS on hypertension not mediated by the biomarker), the natural indirect effect (effect mediated by the biomarker), the total effect (overall PFAS-hypertension association combining direct and indirect effects), and the percent mediated (proportion of the total effect mediated by the biomarker) on the relative-time scale (time ratios to incident hypertension). To mitigate potential selection bias arising from differential participation and follow-up, inverse probability weights were used to re-weight the analytic sample to a pseudo-population representing women at risk of incident hypertension at the time of PFAS assessment in the original SWAN cohort. Weights were estimated using logistic regression models for inclusion in the mediation analytic sample; the denominator model included a priori predictors of both selection and hypertension risk, and the numerator model included an intercept only to stabilize the weights. Details on covariates and weight estimation are provided in the [Supplemental Methods](#).

Sensitivity analyses were performed by repeating the mediation analyses using PFAS and mediators measured at visit 3 (baseline), with follow-up for incident hypertension beginning after visit 3. We also conducted a sensitivity analysis using Cox proportional hazard models instead of AFT, acknowledging the rare outcome assumption.³⁶ For the Cox models, we evaluated the proportional hazards assumption using Schoenfeld residuals. All statistical analyses were performed using R version 4.3.3 (R Foundation for Statistical Computing). Quantile g-computation and causal mediation analyses were conducted using the “qgcomp” (version 2.18.4) and “CMAverse” (version 0.1.0) packages in R.

RESULTS

Table 1 presents the baseline characteristics and inflammatory and hemostatic markers of participants in the SWAN-MPS study. The availability of biomarker measurements and the number of flagged values by study visit are presented in [Supplemental Table 1](#). The median age of the participants was 49 years (IQR: 47, 51). Of 1,342 participants, 682 (50.8%) identified as White, 286 (21.3%) identified as Black, 169 (12.6%) as Chinese, and 205 (15.3%) as Japanese. The baseline median (IQR) concentrations

of inflammatory and hemostatic markers were 1.9 ng/mL (0.6, 5.4) for hs-CRP; 366.4 mg/dL (328.5, 395.8) for fibrinogen; 135.5% (118.8%, 145.5%) for FVIIc; 20.1 ng/mL (10.1, 33.9) for PAI-1; and 7.4 ng/mL (5.3, 9.1) for tPA-antigen. The baseline serum PFAS concentrations, median (IQR), were 1.5 (1.0-2.4) ng/mL for PFHxS, 24.8 (17.6-35.6) ng/mL for total PFOS, 4.1 (2.9-5.8) ng/mL for n-PFOA, 0.6 (0.4-0.8) ng/mL for PFNA, 1.2 (0.7-2.2) ng/mL for EtFOSAA, and 1.5 (0.9-2.3) ng/mL for MeFOSAA. Sb-PFOA, PFDA, PFUnDA, and PFDoDA were not included in further analysis due to their low detection rate, which ranged from 3.7% to 41.3% ([Supplemental Table 2](#)). Because analytic sample sizes differed across biomarkers, we also summarized baseline characteristics within each biomarker-specific analytic subsample in [Supplemental Table 3](#). Baseline characteristics were similar across subsamples, suggesting that differences in biomarker availability and exclusion of flagged values did not substantially alter the composition of the analytic sample.

After adjusting for confounders, all 7 individual PFAS were positively associated with hs-CRP and PAI-1 ([Figure 1](#), [Supplemental Table 4](#)). For hs-CRP, the positive associations for a doubling of serum concentrations ranged from 3.59% (95% CI: 0.01%-7.30%) for Sm-PFOS to 5.32% (95% CI: 1.67%-9.10%) higher for MeFOSAA. Differences in PAI-1 per doubling of PFAS ranged from 2.41% (95% CI: -0.86% to 5.78%) for n-PFOS to 3.04% (95% CI: 0.24%-5.93%) for MeFOSAA. The associations with other hemostatic markers were relatively weak, except positive associations between MeFOSAA and FVIIc activity (0.63%; 95% CI: -0.02% to 1.28%) and PFNA and tPA-antigen (1.49%; 95% CI: -0.01% to 3.01%). Additional adjustment for baseline NSAID usage slightly reduced the associations with hs-CRP while it slightly increased the association with PAI-1 ([Supplemental Figure 2](#), [Supplemental Table 4](#)). However, after applying FDR correction, none of the associations remained statistically significant ([Supplemental Table 5](#)).

Sensitivity analyses showed that the findings were robust to several alternative specifications. PFAS and time interaction analyses did not indicate meaningful variation in associations over follow-up. Restricting the hs-CRP analyses to visits V03-V07 yielded a similar overall pattern of results ([Supplemental Figure 3](#)). Race-by-PFAS interaction results are shown in [Supplemental Figure 4](#). We found no evidence of effect modification by race.

TABLE 1 Participants Characteristics at SWAN Multipollutant Study at Baseline (1999-2000) (N = 1,342)	
Age, y	49.4 (47.4, 51.5)
Race group	
White	682 (50.8)
Black	286 (21.3)
Japanese	169 (12.6)
Chinese	205 (15.3)
SWAN site	
Southeastern Michigan	244 (18.2)
Boston	225 (16.8)
Oakland	293 (21.8)
Los Angeles	362 (27.0)
Pittsburgh	218 (16.2)
Education	
Less than high school	247 (18.4)
High school	428 (31.9)
College	330 (24.6)
Post college	337 (25.1)
Smoking status	
Never smoked	850 (63.3)
Past only	338 (25.2)
Current	154 (11.5)
Passive smoking	
0 person-h	800 (59.6)
1-4 person-h	308 (23.0)
5+ person-h	234 (17.4)
Alcohol consumption	
None	708 (52.8)
≤1 drink/week	324 (24.1)
>1 drink/week	310 (23.1)
Menopausal status	
Premenopause	160 (11.9)
Perimenopause	794 (59.2)
Natural and surgery postmenopause	182 (13.6)
Unknown	206 (15.4)
Nonsteroidal anti-inflammatory drugs use	
No	941 (70.1)
Yes	401 (29.9)
Physical activity score (3-15)	7.8 (6.7,8.9)
Daily calorie intake, Kcal	1,694 (1336,2172)
Body mass index, kg/m2	26.2 (22.6,31.7)
PFAS concentrations (ng/mL)	
n-PFOA	4.1 (2.9,5.8)
PFNA	0.6 (0.4,0.8)
PFHxS	1.5 (1, 2.4)
n-PFOS	17.4 (12.4,24.8)
Sm-PFOS	7.4 (4.7,11)
Total PFOS ^a	24.8 (17.6,35.6)
MeFOSAA	1.5 (0.9,2.3)

Continued in the next column

TABLE 1 Continued	
EtFOSAA	1.2 (0.7,2.2)
Inflammatory and hemostatic markers	
hs-CRP (mg/L)	1.9 (0.6, 5.4)
Fibrinogen (mg/dL)	366.4 (328.5, 395.8)
Factor VIIc (%)	135.5 (118.8, 145.5)
PAI-1 (ng/mL)	20.1 (10.1,33.9)
tPA-antigen (ng/mL)	7.4 (5.3,9.1)
Values are median (IQR) or n (%). ^a Total PFOS was calculated as the sum of n-PFOS and Sm-PFOS. EtFOSAA = 2-(N-ethyl-perfluorooctane sulfonamido) acetic acid; Factor VIIc = coagulation factor VII activity; hs-CRP = high-sensitivity C-reactive protein; MeFOSAA = 2-(N-methyl-perfluorooctane sulfonamido) acetic acid; n-PFOA = linear perfluorooctanoic acid; n-PFOS = linear perfluorooctane sulfonic acid; PAI-1 = plasminogen activator inhibitor-1; PFAS = perfluoroalkyl/polyfluoroalkyl substances; PFHxS = perfluorohexane sulfonic acid; PFNA = perfluorononanoic acid; Sm-PFOS = branched perfluorooctane sulfonic acid; tPA-antigen = tissue-type plasminogen activator antigen.	

The results remained consistent with the primary analysis when flagged extreme values were included (Supplemental Figure 5).

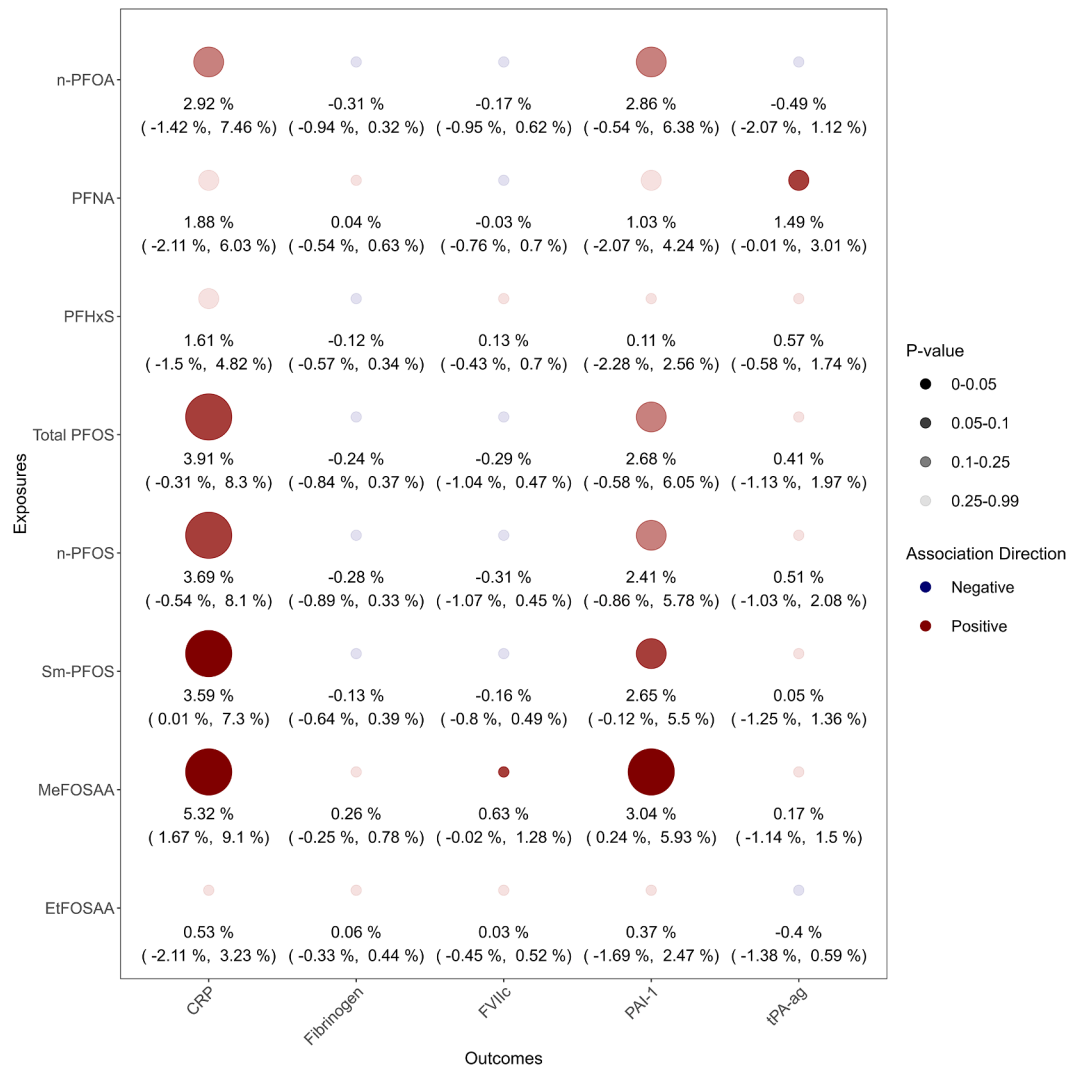
Table 2 presents the qqcomp estimates reflecting associations between PFAS mixtures and 5

hemostatic and inflammatory biomarkers, using the bootstrap method. The joint mixture effects of PFAS when all individual PFAS increased one quartile were 6.15% (95% CI: -0.83% to 13.58%) for hs-CRP and 6.05% (95% CI: 0.29%-12.14%) for PAI-1. No associations were found with other hemostatic markers.

Higher serum PFAS concentrations were generally associated with early onset of hypertension in the causal mediation analyses (Table 3). The total effects combining direct and indirect (mediated) effects ranged from 0.89 to 0.94 except PFHxS with relative survival >1. Across PFAS, natural indirect effects through hs-CRP and PAI-1 were consistently close to 1, indicating limited mediated effects. With hs-CRP as a mediator, percent mediated ranged from 6.2% for PFNA to 15.8% for Sm-PFOS. Mediated effects for PAI-1 were small. Mediation effects were less consistent for other mediators (Supplemental Table 6). Sensitivity analyses confirmed that similar mediation results were observed when using the Cox proportional hazards model (Supplemental Table 7), as well as in models where both PFAS and mediators measured concurrently (Supplemental Table 8). In the Cox models, Schoenfeld residual tests did not show consistent evidence of proportional hazards violations for the primary PFAS exposure terms. Although some covariates showed evidence of nonproportionality in selected models, no additional modifications were made because Cox models were used only as sensitivity analyses and yielded results consistent with the primary AFT models.

DISCUSSION

As shown in the Central Illustration, we evaluated whether inflammatory and hemostatic biomarkers

FIGURE 1 Associations between PFAS Serum Concentrations and Inflammatory/Hemostatic Biomarkers (Model 1)

Percent differences (95% confidence intervals) in inflammatory and hemostatic biomarkers per doubling of serum concentrations of individual PFAS (based on Model 1 which were adjusted for baseline age, baseline BMI, site, ethnicity, degrees, smoking status, baseline menopausal status, alcohol consumption, total activity score, passive smoking status, and total energy intake). Circle size reflects the magnitude of the percent difference, and circle color darkness corresponds to the nominal, unadjusted *P* value (darker shades indicating smaller *P* values). Red indicates positive associations, and blue indicates negative associations. FDR-adjusted *P* values are reported in Supplemental Table 3. CRP = C-reactive protein; EtFOSAA = 2-(N-ethyl-perfluorooctane sulfonamido) acetic acid; FVIIc = coagulation factor VII activity; MeFOSAA = 2-(N-methyl-perfluorooctane sulfonamido) acetic acid; n-PFOA = linear perfluorooctanoic acid; n-PFOS = linear perfluorooctane sulfonic acid; PAI-1 = plasminogen activator inhibitor-1; PFHxS = perfluorohexane sulfonic acid; PFNA = perfluorononanoic acid; Sm-PFOS = branched perfluorooctane sulfonic acid; tPA-antigen = tissue-type plasminogen activator antigen. Total PFOS = n-PFOS + Sm-PFOS.

might represent intermediate pathways linking PFAS exposure to cardiovascular risk. This study examined the association between serum PFAS concentrations and longitudinally measured inflammatory and hemostatic markers in midlife women for up to 16 years of follow-up. We observed nominal positive

associations between select PFAS, particularly PFOS and one of its precursors, MeFOSAA,³⁷ and hs-CRP and PAI-1. These results were directionally consistent with possible links between PFAS exposure, systemic inflammation, and impaired fibrinolysis. However, because none of the individual PFAS-

TABLE 2 Percent Differences (95% CIs) in Inflammatory and Hemostatic Markers When All Individual PFAS^a Increase One Quartile, Calculated Using Quantile G-computation (Qgcomp)

	hs-CRP	Factor VIIc	Fibrinogen	PAI-1	tPA-Antigen
Percent difference (95% CI) ^b	6.15 (−0.83 to 13.58) ^c	1.20 (−0.23 to 2.66) ^c	0.08 (−0.87 to 1.05)	6.05 (0.29–12.14) ^d	1.36 (−1.32 to 4.11)

^aIndividual PFAS included for PFAS mixtures are PFOA (perfluorooctanoic acid), n-PFOS (linear perfluorooctane sulfonic acid), Sm-PFOS (branched perfluorooctane sulfonic acid isomers), PFHxS (perfluorohexane sulfonic acid), PFNA (perfluorononanoic acid), MeFOSAA (2-(N-methyl-perfluorooctane sulfonamido)acetic acid), and EtFOSAA (2-(N-ethyl-perfluorooctane sulfonamido)acetic acid). ^b95% CIs were calculated using cluster-appropriate standard errors based on bootstrapping with 2000 iterations. ^cP < 0.10. ^dP < 0.05.

Factor VIIc = coagulation factor VII activity; hs-CRP = high-sensitivity C-reactive protein; PAI-1 = plasminogen activator inhibitor-1; tPA-antigen = tissue-type plasminogen activator antigen.

biomarker associations remained statistically significant after FDR correction, the findings should be interpreted as suggestive rather than definitive. This study also did not support the mediating role of inflammatory and hemostatic markers in the association between PFAS and hypertension risk.

These findings align with previous epidemiologic studies^{38,39} indicating a positive association between PFAS and CRP levels. A birth cohort study of 738 pregnant women in Denmark found a positive association between serum concentrations of PFOA and PFOS and serum hs-CRP level in mother at pregnancy week 24.³⁸ In a study of 55 osteoarthritis patients from Guangzhou, China, serum PFAS concentrations including PFHxS, PFOS, PFDA, and PFUnDA were positively associated with CRP.¹³

However, other studies, most of which examined pregnant women from maternal-child cohorts with CRP measured at a single time point during pregnancy, failed to observe an association between PFAS and CRP although PFAS concentrations were associated with other inflammatory markers, such as IL-10, IL-6, and tumor necrosis factor-α.^{5,14} With up to 9 repeated measurements over 15 years, our study was able to account for within-participant variability of nonspecific systemic inflammation levels although we were limited to identify specific inflammation pathways reflected by various inflammatory markers.

Our study also found consistent positive associations with PAI-1 levels across various PFAS. Higher PAI-1 levels indicate impaired fibrinolytic function

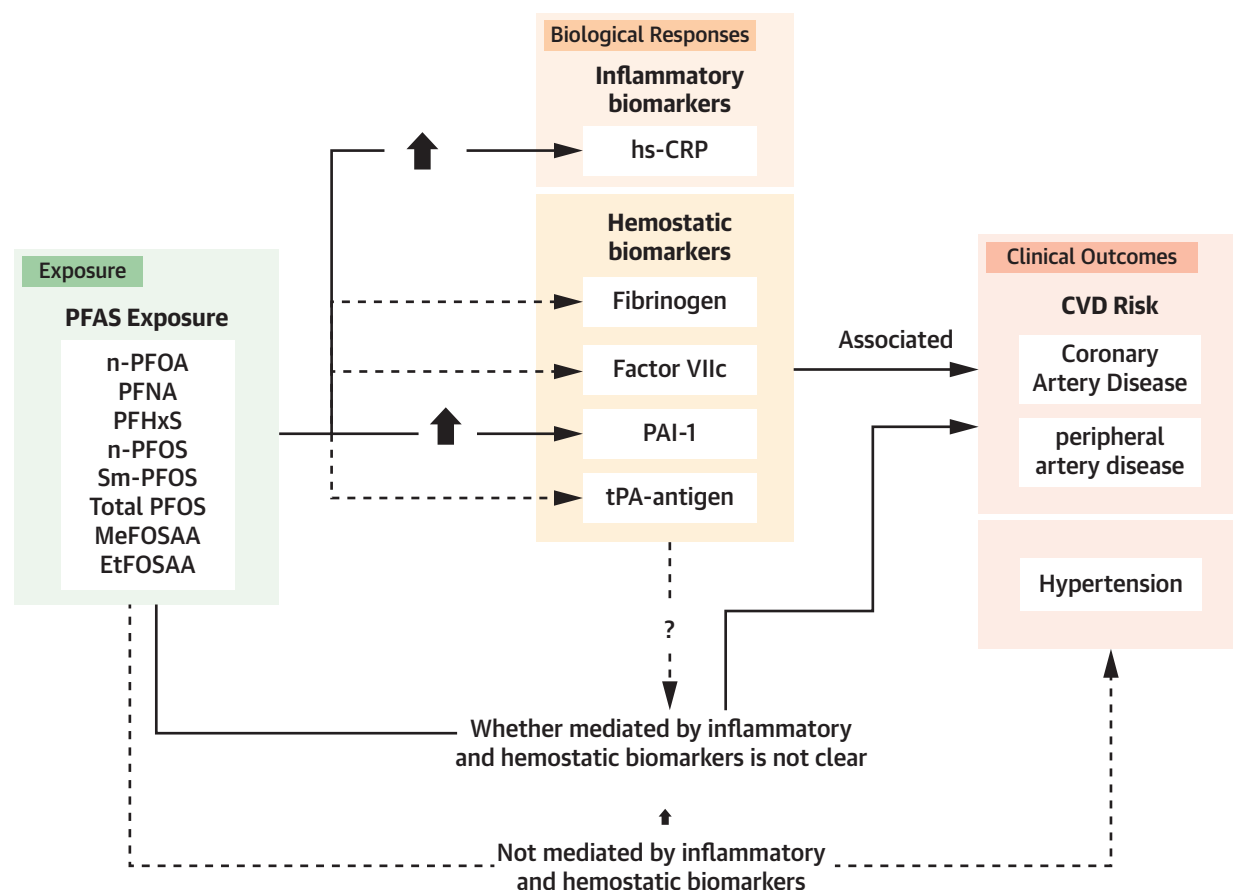
TABLE 3 Results of Causal Mediation Analysis of the PFAS-Hypertension Association Mediated by hs-CRP and PAI-1^{a,b}

Types of PFAS	Mediators ^c	Natural Direct Effect Relative Survival (95% CI)	Natural Indirect Effect Relative Survival (95% CI)	Total Effect Relative Survival (95% CI)	% Mediated
n-PFOA	hs-CRP	0.94 (0.80–1.11)	0.99 (0.96–1.00)	0.93 (0.80–1.08)	14.1% (−199.1 to 201.0)
	PAI-1	0.91 (0.78–1.06)	1.00 (0.97–1.02)	0.90 (0.78–1.05)	3.5% (−100.2 to 90.1)
PFNA	hs-CRP	0.94 (0.81–1.09)	1.00 (0.98–1.01)	0.93 (0.81–1.08)	6.2% (−77.1 to 81.9)
	PAI-1	0.93 (0.80–1.06)	1.00 (0.98–1.01)	0.93 (0.80–1.05)	2.6% (−91.5 to 64.9)
PFHxS	hs-CRP	1.10 (1.00–1.22)	1.00 (0.98–1.01)	1.10 (0.99–1.22)	–
	PAI-1	1.07 (0.96–1.18)	1.00 (0.99–1.00)	1.07 (0.96–1.18)	–
Total PFOS	hs-CRP	0.91 (0.80–1.05)	0.99 (0.97–1.01)	0.90 (0.79–1.04)	8.0% (−76.3 to 103.4)
	PAI-1	0.90 (0.78–1.03)	1.00 (0.97–1.02)	0.89 (0.77–1.02)	0.8% (−53.6 to 59.6)
n-PFOS	hs-CRP	0.91 (0.80–1.06)	0.99 (0.97–1.01)	0.90 (0.79–1.05)	8.0% (−101.8 to 87.9)
	PAI-1	0.90 (0.78–1.03)	1.00 (0.97–1.02)	0.90 (0.77–1.03)	0.7% (−56.9 to 68.7)
Sm-PFOS	hs-CRP	0.94 (0.84–1.07)	0.99 (0.96–1.00)	0.93 (0.83–1.05)	15.8% (−194.2 to 243.1)
	PAI-1	0.93 (0.81–1.05)	0.99 (0.97–1.01)	0.92 (0.81–1.03)	6.0% (−108.0 to 84.6)
MeFOSAA	hs-CRP	0.92 (0.81–1.04)	0.99 (0.97–1.00)	0.91 (0.80–1.03)	7.7% (−67.7 to 127.5)
	PAI-1	0.93 (0.82–1.05)	1.00 (0.98–1.01)	0.92 (0.81–1.04)	5.9% (−75.2 to 86.4)
EtFOSAA	hs-CRP	0.93 (0.85–1.01)	1.00 (0.99–1.01)	0.93 (0.85–1.01)	–
	PAI-1	0.94 (0.86–1.02)	1.00 (0.99–1.00)	0.94 (0.86–1.02)	4.1% (−33.4 to 70.8)

^aCausal Mediation analysis was based on: 1) AFT for the outcome model with hypertension as the dependent variable and both exposure (PFAS) and mediator as independent variables with adjustment for confounders; and 2) linear regression for the mediator model with mediator as the dependent variable and PFAS as independent variable with adjustment for confounders. ^bModels were adjusted for age, age squared, BMI, site, ethnicity, degrees, smoking status, menopausal status, alcohol consumption, total activity score, passive smoking status, and total energy intake at the MPS baseline. ^cMediators including hs-CRP and PAI-1 used measurement from V4, and incidence of hypertension after V4.

AFT = accelerated failure time; EtFOSAA = 2-(N-ethyl-perfluorooctane sulfonamido) acetic acid; hs-CRP = high-sensitivity C-reactive protein; MeFOSAA = 2-(N-methyl-perfluorooctane sulfonamido) acetic acid; n-PFOA = linear perfluorooctanoic acid; n-PFOS = linear perfluorooctane sulfonic acid; PAI-1 = plasminogen activator inhibitor-1; PFHxS = perfluorohexane sulfonic acid; PFNA = perfluorononanoic acid; Sm-PFOS = branched perfluorooctane sulfonic acid. Total PFOS = n-PFOS + Sm-PFOS.

CENTRAL ILLUSTRATION Conceptual Framework Linking PFAS to Cardiovascular Risk



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This central illustration summarizes the hypothesized relationships examined in this study between perfluoroalkyl and polyfluoroalkyl substances (PFAS) exposure and cardiovascular disease (CVD) outcomes. PFAS exposure may influence biological responses reflected by inflammatory biomarkers (hs-CRP) and hemostatic biomarkers (fibrinogen, factor VIIc, plasminogen activator inhibitor-1 [PAI-1], and tissue plasminogen activator antigen [tPA-antigen]). These biomarkers are associated with clinical outcomes, including coronary artery disease, peripheral artery disease, and hypertension. Solid arrows indicate pathways supported by existing evidence and/or evaluated in the current analyses, whereas dashed arrows indicate potential direct effects or mediation pathways that remain uncertain. CRP = C-reactive protein; EtFOSAA = 2-(N-ethyl-perfluorooctane sulfonamido) acetic acid; FVIIc = coagulation factor VII activity; MeFOSAA = 2-(N-methyl-perfluorooctane sulfonamido) acetic acid; n-PFOA = linear perfluorooctanoic acid; n-PFOS = linear perfluorooctane sulfonic acid; PAI-1 = plasminogen activator inhibitor-1; PFHxS = perfluorohexane sulfonic acid; PFNA = perfluorononanoic acid; Sm-PFOS = branched perfluorooctane sulfonic acid; tPA-antigen = tissue-type plasminogen activator antigen. Total PFOS = n-PFOS + Sm-PFOS.

and an elevated risk of blood clot formation, which is linked to hypercoagulation and CVD.⁴⁰ These findings are consistent with the interconnection between PFAS-related inflammation and coagulation, where proinflammatory cytokines upregulate tissue factor and PAI-a and downregulate tPA-antigen, shifting the hemostatic balance toward a prothrombotic state.^{9,12,41,42} On the other hand, PFAS were not associated with other hemostatic markers, such as fibrinogen, FVIIc, and tPA-antigen. These findings suggest that PFAS exposure may specifically

affect pathways associated with the inhibition of fibrinolysis rather than other components of the coagulation system. This specificity could be important in understanding how PFAS are linked to CVD and other health conditions. Further epidemiologic studies are warranted to replicate our findings.

Despite the paucity of human epidemiologic studies on PFAS and hemostasis, our findings align with 2 previous in vitro studies.^{18,19} Toni et al conducted an investigation involving 48 male participants from PFAS-contaminated areas in the Italian

Veneto region and found an increased platelets aggregation among this population compared to 30 low-exposure controls.¹⁸ They reported that PFOA accumulated in platelets, targeted platelets' membrane, and enhanced their membrane fluidity, activation, calcium uptake, and aggregation.¹⁸ These results suggest that high concentrations of PFOA may interfere with the platelet membrane, resulting in altered regulation and functionality of the platelets. Furthermore, exposure to C6O4, a new type of short-chain PFAS detected in surface waters in the Veneto region, was found to significantly increase platelet aggregation and the production of procoagulation platelets micro-particles.¹⁹

Our mediation analysis did not generally support a significant role for hs-CRP or hemostatic markers in mediating the association between PFAS and incident hypertension. In addition, the PFHxS results should be interpreted cautiously. Although PFHxS total-effect estimates were opposite in direction to those of other PFAS in the primary AFT mediation models, the indirect effects were consistently close to the null, and the total-effect estimates were not consistent in Cox sensitivity analyses. Accordingly, we interpret the PFHxS finding as model-sensitive and nonrobust rather than as evidence of a distinct mediator-driven association. One possible explanation is that only single measurements of hs-CRP and hemostatic markers were used in the mediation analyses to ensure that mediator measurements preceded the development of hypertension. Only participants who were free of hypertension at the time of mediator assessment were eligible for mediation analyses, which limited our power to detect mediated effects. In addition, the close timing of measurements for mediators and early cases of hypertension raises the possibility of reverse causation. Established hypertension can promote inflammatory and prothrombotic changes, in which case elevated levels of hs-CRP and PAI-1 may reflect consequences rather than mediators of high blood pressure.⁴³ Importantly, our null mediation findings for hypertension do not exclude hemostatic pathways for other cardiovascular endpoints. Hemostatic dysregulation and thrombosis are more central to the pathogenesis of ischemic events such as myocardial infarction, stroke, and peripheral artery disease than to early blood pressure elevation, where renal and vascular mechanisms may predominate.⁴⁴ Due to the small number of cases in our study, we were unable to evaluate these more clinically relevant CVD outcomes. Future studies with longer follow-up periods may address this question.

Taken together, our results do not establish inflammatory or hemostatic dysregulation as confirmed consequences of PFAS exposure, but they highlight the potential biologic pathways that need further study. Specifically, elevated hs-CRP levels serve as a marker of heightened systemic inflammation, while increased PAI-1 levels reflect impaired fibrinolysis and a pro-thrombotic tendency. Although mechanistic evidence supports the involvement of inflammatory and hemostatic pathways in PFAS-related vascular damage, such as PFAS-induced endothelial dysfunction, oxidative stress, disruption of signaling pathways, and alterations in lipid metabolism,^{45,46} our mediation analysis did not identify hs-CRP or hemostatic markers as significant mediators of the PFAS and hypertension association. These mechanisms are nevertheless thought to play a critical role in vascular damage and the development of conditions such as hypertension, coronary artery disease, strokes, peripheral vascular disease, dyslipidemia, and type-2 diabetes.^{39,46-48} Given the limited epidemiologic studies directly investigating the association between PFAS exposure and incident CVD events, future research should prioritize evaluating the mediating roles of systemic inflammation and hemostatic markers in the PFAS-CVD relationship.

This study is the first to investigate the association between PFAS and both hs-CRP and hemostatic biomarkers measured longitudinally. Although previous studies have shown variable associations between PFAS and CRP levels, leading to inconsistent conclusions, the longitudinal design and large sample size in this study enhance the robustness of our findings, offering clearer insights into the association between PFAS and inflammatory and hemostatic responses compared to earlier cross-sectional studies.

STUDY LIMITATIONS. This study also has several limitations. First, our study population included only women who were middle-aged at the commencement of the study. Thus, our findings may not be generalizable to men or people of other age groups. Future studies should include broader demographics to validate these findings across the general population. Second, serum PFAS concentrations were measured only at a single time point at baseline, which may result in exposure misclassification. However, the long-alkyl-chain PFAS included in the present study have long biological half-lives.⁴⁹ Our previous study using a subset of 75 women from SWAN showed the estimated halving time ranged from 3.7 years for Sm-PFOS to 11.2 years for n-PFOA,²⁶

suggesting that serum PFAS concentrations used in this study could be reliable exposure indicators over the study period. Another SWAN study found longitudinal PFAS measurements from 1999 to 2011 demonstrated declines in legacy PFAS, alongside increases in PFNA.⁵⁰ But this misclassification would be expected to attenuate associations toward the null rather than create positive findings. Third, our findings, particularly the observed associations between PFAS and hs-CRP and PAI-1 should be interpreted with caution given the lack of statistical significance after FDR correction. The observed percent differences per doubling of PFAS were modest, and their clinical significance remains uncertain. Although repeated biomarker measurements and longitudinal mixed-effects models helped account for within-person variability, small changes in hs-CRP and PAI-1 may be difficult to distinguish from normal biological fluctuation or assay-related variability. Fourth, although we adjusted for a wide range of potential confounders, we cannot rule out residual confounding that may be related to both PFAS and inflammation or hemostasis, such as occupation and factors that influence serum PFAS concentrations (eg, bile acids,⁵¹ renal function⁵²). In particular, impaired renal clearance may influence circulating PFAS concentrations and may also be associated with inflammatory and hemostatic dysregulation.⁵² Future studies with repeated measures of renal function and metabolic health are needed to better distinguish these pathways.

CONCLUSIONS

Both individual PFAS and PFAS mixtures showed suggestive patterns of associations with systemic inflammation and thrombotic risks, but these associations did not remain statistically significant after FDR correction. These findings highlight the need for further research to evaluate the potential cardiovascular implications of PFAS exposure. Future studies may focus on the mediating effects of inflammatory and hemostatic biomarkers in the association between PFAS and clinical CVD events. As PFAS are modifiable risk factors, strategies to minimize exposure are of public health relevance. Clinicians may wish to remain aware of PFAS as potential, yet often overlooked, risk factors for CVD and be prepared to counsel patients about sources of exposure and possible health effects.

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APPENDIX For an expanded Methods section as well as supplemental tables and figures, please see the online version of this paper.